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MILITARY SPECIFICATION

LINER, GROUND TROOP'S - PARACHUTIST'S HELMET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers ballistic protective helmet liners, designed for use with Helmet, Ground Troop's - Parachutist's, M-1, or separately as a head covering and consisting of a laminated coated nylon fabric liner shell with liner suspension assembly, metal fittings and a detachable headband and neckband. A chin strap suspension assembly and chin strap are provided with the type II parachutist liner.

1.2 Classification. Helmet liners shall be of the following types as specified (see 6.2).

Type I - Liner, Helmet (Ground Troop's)
Type II - Liner, Helmet (Parachutist's)

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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SPECIFICATIONS

FEDERAL

V-T-285	- Thread, Polyester
FF-R-556	- Rivet, Solid, Small; Rivet, Split, Small; Rivet, Tubular, Small; Flat Washer (Burr); and Cap, Rivet; General Purpose
UU-P-268	- Paper, Kraft, Wrapping
DDD-L-20	- Label, For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-1797	- Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

MILITARY

MIL-F-495	- Finish, Chemical, Black, for Copper Alloys
MIL-H-1988	- Helmet, Ground Troop's - Parachutist's M-1
MIL-F-10884	- Fasteners, Snap
MIL-C-12369	- Cloth, Ballistic, Nylon
MIL-H-41802	- Headband and Neckband, Ground Troops'- Parachutists' Helmet Liner
MIL-H-43178	- Hardware; for Helmets, Helmet Liners, and Bands
MIL-S-43355	- Strap, Chin; and Suspension Assembly, Chin Strap
MIL-C-43460	- Coatings, Lusterless, For Helmets and Liners
MIL-T-43566	- Tape, Textile, Cotton, General Purpose, Natural or in Colors
MIL-S-43620	- Suspension Assembly, Ground Troops' Helmet Liner
MIL-P-46593	- Projectile, Calibers .22, .30, .50 and 20 MM Fragment-Simulating

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-662	- Ballistic Test for Armor
MS-51936-6	- Washer, Flat (Burr)

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DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

8-2-61	- Liner, Ground Troop's - Parachutist's Helmet; Shell Contour and Dimensions for Type I and II
8-2-62	- Liner, Ground Troop's - Parachutist's Helmet; Assembly and Details Types I and II
8-2-64	- Liner, Ground Troop's - Parachutist's Helmet; Preform
8-2-581	- Liner, Ground Troop's - Parachutist's Helmet; Assembly and Details; Type II Parachutist's
8-2-584	- Liner, Ground Troop's - Parachutist's Helmet; Attachment of Chin Straps and Chin Strap Suspension Assembly to Type II Liner

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

G 23 - Operating Light-Exposure Apparatus (Carbon-Arc Type) With and Without Water for Exposure of Nonmetallic Materials

D 822 - Operating Light and Water-Exposure Apparatus (Carbon-Arc Type) for Testing Paint, Varnish, Lacquer, and Related Products

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide samples. Guide samples of the helmet liner or components, when furnished, are solely for guidance and information to the contractor (see 6.5). Variation from this document may appear in the samples, in which case this document shall govern.

3.1.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.1.3 Production sample. Unless otherwise specified (see 6.2), one helmet liner shell from each mold that the contractor proposes to use in production shall be submitted to the contracting officer within one week after the mold is placed into operation. The production sample shall be inspected as specified in 4.3.1.

3.1.4 Standard panel for color, gloss and texture of helmet liner. The standard sample for use as reference panels for color, gloss, and texture shall be submitted to the contracting officer as specified in MIL-C-43460.

* 3.2 Materials and components. Recycled material may be used for nontextile components (see 6.4).

3.2.1 Liner shell materials.

3.2.1.1 Reinforcing material. The reinforcing material of the shell shall be nylon ballistic cloth conforming to class 1 of MIL-C-12369. The cloth used for the inner and outer surface plies of the shell shall be dyed olive green. The cloth used for all other plies need not be dyed.

3.2.1.2 Laminating resin. The resin for coating and laminating the reinforcing material shall be a catalyzed system composed of phenol formaldehyde and polyvinyl butyral resins (see 4.4.1.1).

3.2.1.3 Protective coating. The protective coating for the shell shall conform to type II, class A of MIL-C-43460.

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3.2.2 Suspension assembly, type I liner. The suspension assembly for the type I liner shall conform to MIL-S-43620.

3.2.3 Headband and neckband, types I and II liners. When specified (see 6.2), the types I and II liners shall be furnished with a headband and a neckband. The headband shall conform to type I of MIL-H-41802 and the neckband shall conform to type II, class 2 of MIL-H-41802.

3.2.4 Chin strap, type II liner. The type II liner shall be furnished with a chin strap conforming to type I of MIL-S-43355.

3.2.5 Chin strap suspension assembly, type II liner. The type II liner shall be furnished with a chin strap suspension assembly conforming to type II of MIL-S-43355.

3.2.6 Snap fasteners for type II liners. Snap fasteners (button and socket only) for the type II liner shall conform to style 2A, finish 2 of MIL-F-10884. The length of the snap fastener eyelet shall be sufficient to provide a rolled or peened edge of 1/32 inch minimum.

3.2.7 Rivets. All rivets for the types I and II liners except for attaching the studs on the type I liner shall be tubular, oval head, in accordance with type XII, class 3, grade F, 0.146 nominal diameter of FF-R-556. The length shall be sufficient to permit a rolled or peened 1/32 inch edge and hold the item securely. The rivets for attaching the studs on the type I liner shall be as shown on Drawing 8-2-62. All rivets shall be given a black chemical finish conforming to MIL-F-495.

3.2.8 Washers. The washers (burrs) used in the attachment of the chin strap suspension assembly to the type II liner shall be brass conforming to FF-R-556 and MS 51936-6.

3.2.9 Buckle and clip assemblies, neckband. The buckle and clip assemblies for attaching the neckband shall conform to type III of MIL-H-43178.

3.2.10 Studs. The studs for attaching the suspension assembly on the type I liner shall be as shown on Drawing 8-2-62. The stud shall be fabricated from type 302 corrosion-resisting steel in annealed condition.

3.2.11 "A" washer. The "A" washer for attaching the suspension assembly to the type II liner shall conform to type X of MIL-H-43178.

3.2.12 Eyelet. The eyelets for the suspension assembly on the type II liner shall conform to the requirements specified on Drawing 8-2-581. The eyelets shall be given a black chemical finish conforming to MIL-F-495.

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3.2.13 Buckle, bar. The bar buckle for the suspension assembly on the type II liner shall conform to type I of MIL-H-43178.

3.2.14 Thread, polyester. The thread for all stitching on the suspension assembly for the type II liner shall conform to types I or II, class 1, subclass A, size E, olive drab S-1 (C.A. 66022) of V-T-285. The dyed thread shall show fastness to perspiration and light equal to or better than the standard sample (see 6.5).

3.2.15 Tape, cotton. The cotton tape for the suspension assembly for the type II liner shall conform to type I, class 3, 1 1/8 inches wide, olive drab 7 of MIL-T-43566.

3.3 Design.

3.3.1 Liner (type I). The type I helmet liner shall consist of a laminated nylon fabric liner shell with removable suspension assembly, all hardware components shown on Drawing 8-2-62 and, when specified, a headband and neckband (see 6.2).

3.3.2 Liner (type II). The type II helmet liner shall consist of a laminated nylon fabric shell with a nonremovable suspension assembly, chin strap, chin strap suspension assembly, all hardware components shown on Drawing 8-2-581 and, when specified, a headband and neckband (see 6.2).

3.4 Construction. The construction of the helmet liner and assembly of components to the helmet liner shall conform in all respects to the applicable drawings listed in 2.1 and as specified herein.

3.4.1 Stitching, machine. All stitching on the suspension assembly for the type II liner, except bartacking, shall conform to FED-STD-751, type 301 with 6 to 8 stitches per inch.

3.4.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/2 inch minimum. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.4.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch in back of the end of the stitching.

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b. Thread breaks, two or more consecutive skipped or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continued over the defective area, and continued a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials and restitching in the required manner.

c. When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.1.2 Automatic stitching. Automatic machines may be used to perform any of the required stitch patterns, provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met, and at least three tying, overlapping, or back stitches are used to secure the ends of the stitching.

3.4.1.3 Bartacking. Bartacks, if used, shall be 1 ± 1/16 inch long and 1/8 ± 1/32 inch wide and shall contain 42 stitches per bartack. Bartacking shall be free of thread breaks and loose stitching.

3.4.1.4 Thread ends. All thread ends shall be trimmed to a length of not more than 1/4 inch.

* 3.4.2 Shell liner. The liner shell shall be a laminated structure composed of resin coated reinforcing material bonded by heat and pressure (see 6.9). The reinforcing material specified in 3.2.1.1 shall be coated on both sides with equal amounts of the laminating resin specified in 3.2.1.2 (see 4.4.1.1). The outer material shall be olive green color (inside and outside surfaces of the helmet liner). The resin content of the reinforcing material shall be 18 to 21 percent solids based on a reinforcing material weight of 14.0 ounces per square yard. The preforms (see 6.8), shall be assembled so that the resulting helmet liner shall have not less than four layers of fabric throughout, nor shall it have fabric gaps wider than 3/32 inch or longer than 2 inches in the inside and outside surfaces of the finished shell, nor shall there be any pleat, wrinkle or crease that is longer than 1/2 inch. Any pleat, wrinkle, or crease 1/2 inch or less in length shall be smooth. The outer and inner surfaces of the molded shell shall be smooth. Both the inside and outside surfaces shall be free from any hole, void, delamination, pimple, blister, cracking, crazing, dry spot, area of nonresin flow, resin streak, and also any pit greater than 1/8 inch in diameter and the depth of one ply. Permissible gaps and pits shall be filled with epoxy resin in accordance with 3.5.1. The liner shall be formed during a single molding cycle, which is

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defined as the time between the closing and opening of the mold one time only (breathing or degassing is not considered opening or closing the mold in this context). The shell shall not be remolded after this single molding cycle. The contractor is responsible for 100 percent inspection immediately after the molding cycle (see 4.4.2.1), and before further processing is undertaken, for delamination and blisters. Delaminations and blisters, as well as evidence of delaminations and blisters, are not acceptable; neither is any shell that contains a delamination or blister as a result of the molding operation and was modified by any means prior to finishing. Metallic fastening devices, such as staples, are not permitted in the preform unless they are in the trim area. Spot welding or sewing with thread shall be permitted provided that evidence of these procedures are not visible on the outside surface of the finished liner. Holes for rivets and snap fasteners shall be located as shown on Drawing 8-2-61. There shall be no pleat, wrinkle or crease on the inside or outside of the helmet liner, except as indicated above, nor shall there be any dents, punctures or tears in the shell material resulting from attachment of hardware components.

* 3.4.2.1 Liner shell dimensions. The overall dimensions of the liner shell after molding, with edges burnished (fused) and holes drilled or punched, but prior to subsequent manufacturing operations shall be as specified in table I. The examination for length, height, and width shall be made on a Government furnished gage (see 3.10).

TABLE I. Shell dimensions

Dimension	Measurement (inches)
Thickness 1/	0.087 minimum
Width at widest point unrestrained	8 21/32 plus or minus 1/8
Length (width restrained to 8 21/32 inches)	10 21/32 plus or minus 1/8
Height (width restrained to 8 12/32 inches)	6 7/16 plus or minus 1/8

1/ One measurement shall be made in each of the following locations: at the front, at the back, or on each side, and at the crown. Measurements shall be made both above and below the plane of the suspension assembly rivet holes.

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3.4.3 Installation of suspension assembly into type I helmet shell. The suspension assembly shall be installed in the type I helmet liner as shown on Drawing 8-2-61. Each clip shall be fully seated over the stud. The clips may be installed by hand or mechanically and shall meet the performance requirement specified in 3.6.6. If the clips are installed by a mechanical method, no damage shall result to the clips or studs nor shall the studs or rivets in the helmet liner become loosened.

3.4.4 Setting of rivets and snap fasteners. Rivets and snap fasteners shall be securely peened or set to the degree that there is not looseness between the liner and the components being attached; nor shall the rivets or snap fasteners be peened or set too tightly to the degree that the structure of the liner is crushed. All burrs on rivets shall be removed.

3.4.5 Settings of eyelets. Holes for setting eyelets in the suspension assembly for the type II liner shall be as specified on Drawing 8-2-581. The eyelets shall be securely clinched without splitting and in a manner that will prevent detachment from or cutting of the adjacent material.

3.4.6 Repairs. Except as otherwise specified herein, repairs are not allowed to be made to the helmet liner and its components.

3.4.7 Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.3 shall be removed from production and replaced with nondefective and properly matched components.

3.5 Finish.

3.5.1 Surface preparation. Outside surface preparation of the liner shell shall conform to type II of MIL-C-43460.

3.5.2 Edge finish. The periphery of the helmet liner at the pinch-off line shall be trimmed and burnished (fused) to eliminate sharp edges and to provide a uniformly smooth, sealed edge.

3.5.3 Coating. After the surface preparation specified in 3.5.1 the entire outside surface of the helmet liner shall be coated with the protective coating specified in 3.2.1.3. The coated liner shall be in conformance with all of the requirements applicable to type II, class A of MIL-C-43460. The coating shall not furrow, flake, or peel when scratched with the fingernails.

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3.6 Performance.

3.6.1 Impact resistance. The finished liner equipped with liner suspension assembly, headband, and neckband shall withstand an impact force of 40 foot pounds when tested as specified in 4.4.3.3. There shall be no fracture or delamination of the liner, pulling out of any rivet, stud, eyelet or clip, or breakage of any rivet, stud, eyelet, clip or buckle. There shall be no tearing away of the tape at the rivets or tearing of any tape more than half-way across the width of the tape except when tearing occurs at the teeth of the cradle adjustment buckle. The exterior finish shall show no flaking, peeling, loss of adhesion or other failure of the finish except in the immediate, imprinted area of impact.

3.6.2 Accelerated weathering resistance. The exterior finish of the liner shall show no more than slight chalking or color change when tested for accelerated weathering resistance as specified in 4.4.3.3, nor shall the finish show flaking or peeling.

* 3.6.3 Accelerated aging. When the finished liner is tested as specified in 4.4.3.3 there shall be no evidence of blisters or delamination or surface irregularities such as flat areas, bumps, or depressions, nor shall the peripheral edge be scalloped or rippled. In addition, the liner shall fit within an M-1 helmet conforming to MIL-H-1988 with not more than 1/4 inch protruding above the edge of the M-1 helmet at any point around the periphery.

3.6.4 Water immersion. The exterior coating of the helmet liner shall show no evidence of softening, peeling, or blistering when tested as specified in 4.4.3.3.

3.6.5 Ballistic resistance. The average V_{50} ballistic limit for each lot of helmet liners shall be not less than 830 feet per second when tested as specified in 4.4.3.3. The V_{50} ballistic limit for each sample unit liner shall be not less than 800 feet per second.

3.6.6 Clip to stud attachment type I liner. Each clip of the suspension assembly for the type I liner shall be capable of being installed in the liner by hand when tested as specified in 4.4.3.3.

3.7 Weight. The weight of the finished type I liner (see 3.3.1) shall not exceed 15 ounces. The weight of the finished type II liner (see 3.3.2) shall not exceed 15 1/2 ounces. The weight shall be taken without the headband and neckband.

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3.8 Identification marking.

3.8.1 Type I liner. The type I liner shall be marked for identification, using a black indelible marking ink (see 4.4.1.1). The markings shall be located as shown on Drawing 8-2-62. The identification markings shall consist of the information listed below. Size of characters for the nomenclature shall be 3/16 inch minimum and for all other markings 1/8 inch minimum. The markings shall be clearly legible with no smearing, bleeding or offsetting.

LINER, HELMET, GROUND TROOP'S, TYPE I

Stock Number
Contract Number

3.8.2 Type II liner. The type II liner shall be marked for identification in the location shown on Drawing 8-2-581 and shall conform to type IV, class 5 of DDD-L-20.

3.9 Molded-in marking. Each liner shall be marked in the inside crown area with the contractor's mark or symbol and the mold number. The marking shall be accomplished by "mold-in" process.

3.10 Government loaned property. The contracting officer will furnish the contractor one height, width, and length gage, for use during the duration of the contract. The gage shall be returned to the contracting officer at the completion of the contract or purchase order.

3.11 Workmanship. The finished liners shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the specified acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3) and production sample inspection (see 4.3.1).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), the preproduction sample submitted in accordance with 3.1.2 shall be inspected for all the provisions of this document.

4.3.1 Production sample inspection. The production sample referenced in 3.1.3 shall be visually and dimensionally examined to determine that the peripheral contour of the liner is symmetrical and undistorted within the defective limitations specified in 4.4.3.2.1.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance with each shipment or lot certifying the following:

a. That the laminating resin used is a catalyzed system composed of phenol formaldehyde and polyvinyl butyral resins (see 3.2.1.2).

b. That the distribution and amount of resin coating meets the requirements specified in 3.4.2.

c. That the compound for dipping the ends of the tape on the suspension assembly for the type II liner is as specified on Drawing 8-2-581.

d. That the marking ink used on the type I liner is indelible.

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4.4.2 In-process inspection. In-process inspection shall be performed in accordance with the examinations specified in 4.4.2.1, 4.4.2.2, 4.4.2.3, 4.4.2.4, and 4.4.2.5. Unless otherwise specified, defects are to be scored on an individual basis (e.g., each dimension, each wrinkle, each pit, each blister).

4.4.2.1 Visual examination of liner shells prior to trimming and other finishing operations. The liner shells prior to trimming and other finishing operations shall be one hundred percent examined for any evidence of a blister or delamination. Any blister or delamination, as well as any evidence of a blister or delamination that was modified by any means after molding (see 3.4.2), shall be classified as a defect and the liner shall be rejected.

4.4.2.2 Dimensional examination of liner shells prior to application of protective coating. The liner shells after molding, with edges burnished (fused) and holes drilled or punched, but prior to subsequent operations, shall be examined for hole size and locations indicated on Drawing 8-2-61 and also the dimensions indicated in table I. The lot size shall be expressed in units of molded liner shells, and the sample unit shall be one molded liner shell. Any dimension not within the specified tolerance shall be classified as a defect. The inspection level shall be S-3 and the acceptable quality level (AQL) shall be 4.0, expressed in terms of defects per hundred units.

* 4.4.2.3 Visual examination of liner shells prior to application of protective coating. Prior to application of the protective coating, the liner shells shall be examined for visual defects. The defects shall be classified in accordance with table II. The lot size shall be expressed in units of molded liner shells, and the sample unit shall be one liner shell. The inspection level shall be II, and the AQL shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE II. Classification of liner shell defects

Examine	Defect	Classification	
		Major	Minor
Liner shell (exterior and interior)	Any nylon fibers visibly cut or raised on the shell body	101	
	Any surface dent, depression, or area not smooth:		
	- seriously affecting appearance or serviceability	102	
	- affecting appearance or serviceability but not seriously		201
	Uneven, sharp, or jagged edge:		
	- seriously affecting appearance or serviceability	103	
	- affecting appearance or serviceability but not seriously		202
	Edge not uniformly smooth (other than sharp, uneven or jagged)		203
	Edge not sealed	104	
	Any delamination, pimple, or blister	105	
	Any evidence of cracking or crazing	106	
	Any evidence of dry spot, any area or resin streak (except for permissible repairs) or nonresin flow, or any other molding deficiency	107	
	Any cloth gap or pit, except those specified as permissible (see 3.4.2)	108	
	Any void 1/	109	
	Any pleat, wrinkle, or crease longer than 1/2 inch but less than 1 inch		204
	Any pleat, wrinkle, or crease 1/2 inch or less in length that is not smooth		205
	Any pleat, wrinkle, or crease 1 inch or longer	110	

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TABLE II. Classification of liner shell defects - Continued

Examine	Defect	Classification	
		Major	Minor
Liner shell (exterior and interior) -Continued	Any permissible gap or pit not resin filled as specified (see 3.4.2)	111	
	Any unauthorized repair or reworking		206
	Any evidence of metallic fasteners	112	
	Color of surface plies not olive green		207

1/ A void can also occur along the periphery of the liner edge caused by short plies or the creeping up of a preform, thus exposing an inner ply.

* 4.4.2.4 Visual examination of suspension assembly for type II liner prior to installation into helmet liner. The suspension assembly for the type II liner shall be examined for visual defects prior to installation into the helmet liner. The defects shall be classified in accordance with table III. The lot size shall be expressed in units of suspension assemblies, and the sample unit shall be one suspension assembly. The inspection level shall be II, and the AQL shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE III. Classification of suspension assembly defects

Examine	Defect	Classification	
		Major	Minor
Tape	Any hole, cut, or tear	114	
	Frayed edge, not firmly or tightly woven		209

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TABLE III. Classification of suspension assembly defects - Continued

Examine	Defect	Classification	
		Major	Minor
Tape - Continued	End of tape not pinked or not dipped as specified		210
	Color not as specified		211
Buckle	Broken or malformed, corroded area, burr, or sharp edge	115	
	Not attached in orientation specified (i.e., buckle reversed or center bar does not face in specified direction)	116	
Eyelet	Broken or malformed	117	
	Clinched excessively tight, cutting adjacent material	118	
	Insecurely clinched to a degree that eyelet may be detached from material	119	
Seams and stitchings:			
Open seams	Up to and including 1/2 inch		212
	More than 1/2 inch	120	
NOTE:	A seam shall be classified as an open seam when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.		
Stitch type	Incorrect stitch type	121	
Stitch tension	Loose tension, resulting in a loose top or bobbin thread		213
	Excessively tight tension, resulting in puckering of material		214

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TABLE III. Classification of suspension assembly defects - Continued

Examine	Defect	Classification	
		Major	Minor
Seams and stitchings - Continued			
Stitches per inch	One stitch less than minimum specified		215
	Two or more stitches less than minimum specified	122	
	One or more stitches in excess of maximum specified		216
Stitching ends (on stitch type 301)	Not secured as specified		217
Thread breaks, skipped or runoff stitches (on type 301 stitching)	Thread breaks, or two or more consecutive skipped or runoff stitches overstitched less than 1/2 inch in each direction beyond the defective stitching area		218
NOTE:	Thread breaks, or two or more consecutive skipped or runoff stitches not overstitched shall be classified as open seams.		
Bartacks (if used)	Loose stitching, incomplete or broken		219
	Any omitted	123	
Cleanness	Spots or stains clearly noticeable		220
	Thread ends not trimmed specified		221

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4.4.2.5 Dimension examination of suspension assembly for type II liner prior to installation into helmet liner. The suspension assembly for the type II liner shall be examined for dimensional defects prior to installation into the helmet liner. The examination shall be for all dimensions on the drawings, including stitch margins, that can be examined on the completed suspension assembly. Any dimension exceeding the applicable tolerance shall constitute a defect. The lot shall be expressed in units of suspension assemblies, and the sample unit shall be one suspension assembly. The inspection level shall be S-3, and the AQL shall be 4.0 (one class), expressed in terms of defects per hundred units.

4.4.3 Inspection of the end item.

4.4.3.1 Examination of components. The following listed components shall be inspected against the quality assurance provisions of the applicable documents prior to assembly or packaging, as applicable, with the helmet liner.

Component	Document
Suspension assembly, type I liner	MIL-S-43620
Headband and neckband, types I and II liners	MIL-H-41802
Chin strap, type II liner	MIL-S-43355
Chin strap suspension assembly, type II liner	MIL-S-43355

4.4.3.2 Examination of the end item. The defects found during the examination shall be classified in accordance with 4.4.3.2.1 and 4.4.3.2.2. The lot size shall be expressed in units of complete helmet liners. The sample unit shall be one complete type I or type II helmet liner complete with all applicable components except that the chin strap (type II liner), headband and neckband, when required, which are packaged separate from the helmet liners, need not be considered as part of the sample unit. Unless otherwise specified, defects are to be scored on an individual basis (e.g., each area of no coating, each defective rivet, each defective "A" washer).

* 4.4.3.2.1 Visual examination. The end item shall be inspected for the defects listed in table IV. The inspection level shall be II, and the AQL shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE IV.

Examine	Defect	Classification		Inued fication
		Major	Minor	
	accordance			
	ts (unless			
	n):			
	pearance or			
		125		
	r service-			
	sly		232	
		126		
	placed,			
	properly			
		127		
	ffixed or			
		128		
	, or malformed	129		
	ing properly:			Minor
	pearance or			
		130		
	r service-			
	ly		233	215
		131		
	Any read			
	or brui			
	Any ridg			216
	or blis			
	Color, g			217
	Coating			
	scratch			
	Grease o			
	matter			
	Paint sm			
	Any char			218
	with sp			
	otherwi			
	- seri			
	serv			
	- affe			
	abil			
	visual defects			219
	Any comp			
	Any comp			
	operati			220
	perform			221
	Any comp			
	not fir			
	Any comp			
	Any comp			
	- seri			
	serv			
	- affe			
	abil			
	Any unaut			
	no coating,		223	
	hrough)		224	
			225	
	ratch		226	
	g, wrinkle,		227	
	ot uniform		228	
	r peels when			
	nail		229	
	er foreign		230	

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TABLE IV. Classification of visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Liner	Any obvious distortion or nonsymmetry	132	
	NOTE: For liners on which the degree of distortion or nonsymmetry is questionable as to whether a defect should be assessed, the following methods shall be used to determine conformance. (1) Obvious distortion. Draw a symmetrical outline of the liner periphery in accordance with Drawing 8-2-61. Line up the front and back benchmarks and one side benchmark of the liner with the appropriate benchmarks on the outline. Draw the outline of the liner. The second outline shall not deviate more than 1/4 inch from the first outline at any point. (2) Nonsymmetry. The drawing of the periphery of the liner in question shall be folded along the longitudinal axis. The outline of the halves of the liner shall not deviate more than 3/8 inch from each other at any point.		
	Any evidence of blister or delamination	133	
	Any blister or delamination that has been reworked	134	

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TABLE IV. Classification of visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Interior and exterior	Any hole	135	
	Any unauthorized repair or reworking	136	
	Any condition that affects outside appearance:		
	- seriously affecting appearance or serviceability	137	
	- affecting appearance or serviceability but not seriously		234
Hardware (general)	Any sharp edge or burr		235
	Any rivet or eyelet part of snap fastener not properly peened or set but will adequately secure the components together and is not expected to become detached during use		236
	Any rivet or eyelet part of snap fastener too short or not adequately or properly peened or set to a degree where it may become detached from the assembly during use	138	
	Any rivet or snap fastener affixed too tightly resulting in cutting or crushing of shell		237
	Buckle and clip assembly not attached in orientation specified	139	
	Any "A" washer attached with base turned more than 1/8 inch from horizontal edge of peripheral band (type II liner)		238

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TABLE IV. Classification of visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Suspension assembly (type II liner)	Peripheral band not a taut fit between any two attachment points on liner shell	140	
	Not attached in correct orientation	141	
Identification markings	Omitted, illegible, incorrect, incomplete, or misplaced		239
Molded-in marking	Omitted or illegible		240

4.4.3.2.2 Weight examination. The finished liner shall be examined for defects in weight. Any type I liner weighing more than 15 ounces or any type II liner weighing more than 15 1/2 ounces shall be classified as a defect. The inspection level shall be S-3, and the AQL shall be 4.0, expressed in terms of defects per hundred units.

* 4.4.3.3 Testing of the end item. The end item shall be tested for the characteristics listed in table V. The lot size shall be expressed in units of helmet liners. The sample unit for each test shall be one helmet liner. The inspection level shall be S-2 except for operation of removable suspension assembly, which shall be S-4. Except for ballistic resistance, the results shall be reported as pass or fail, and the AQL for each characteristic shall be 2.5, expressed in terms of defects per hundred units. For ballistic resistance, the test results shall be individually reported for each sample unit as the average of all applicable impacts conducted on the sample units. The V_{50} limit for the lot shall be reported as the average of all liners tested. Failure of any individual sample unit or lot average shall be cause for rejection of the lot.

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TABLE V. Testing of the end item

Characteristic	Document reference: Requirement	Document reference: Test method	Requirements applicable to: Sample unit	Requirements applicable to: Lot average	Number determinations per sample unit
Impact resistance	3.6.1	4.5.1	X	-	1
Accelerated weathering resistance	3.6.2	ASTM G 23 and/or D 822 1/, 2/	X	-	1
Accelerated aging test	3.6.3	4.5.2	X	-	2
Water immersion	3.6.4	4.5.4	X	-	1
Operation of removable suspension assembly, type I liner	3.6.6	4.5.5	X	-	1
Ballistic resistance	3.6.5	4.5.3	X	X	1

1/ The test panel shall consist of a 2 by 4 inch specimen of the helmet liner. The time of exposure shall be 200 hours.

2/ This test shall be performed only on the first lot of liners in each contract, or if the contractor has changed his protective coating material or the source of this material during the contract.

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4.4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table VI. The lot size shall be the number of shipping containers in the inspection lot. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL shall be 2.5, expressed in terms of defects per hundred units.

TABLE VI. Packaging inspection

Examine	Defect
Markings, shipping containers	Omitted, incorrect, illegible, or of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number of liners or components per container is more or less than required

4.4.5 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with table VII. The lot size shall be the number of palletized unit loads in the inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1, and the AQL shall be 6.5, expressed in terms of defects per hundred units.

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TABLE VII. Palletization characteristics

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, or of improper size, location, sequence or method of application

4.5 Tests.

4.5.1 Impact resistance.

4.5.1.1 Test apparatus. Standard AML wooden head form, size large (see 6.6). Eight-pound iron ball of approximately 4 inches diameter. A fixed ball release apparatus equipped with an electromagnet for releasing the 8-pound iron ball.

4.5.1.2 Preparation of helmet liner for testing. In order to establish a clearance between the crown of the liner to be tested and the head form, adjustments shall be made. With the liner suspension assembly attached, pull center head suspension cradle strap of the shell assembly to the loop stitching line or pull it taut, whichever occurs first. Adjust cross straps to equal of center strap. Mark each strap at the point where it enters the buckle. Release all straps until the mark is 1 1/2 inches from the original point of marking. These measurements are to be made while holding each strap taut.

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4.5.1.3 Test procedure. With the headband attached, mount the adjusted liner specified in 4.5.1.2, on the head form in a position similar to that in which it would be worn on a man's head. The 8-pound ball shall be dropped vertically from a height of 5 feet (measured from the bottom of the ball to the crown of the liner) to impact the center of the helmet liner crown. Examine the liner for conformance with the requirements specified in 3.6.1. The headband shall be adjusted on the head form 5 1/2 inches from the base to the upper edge of the headband.

4.5.2 Accelerated aging test. The test specimen shall be subjected to saturated steam for a period of 30 minutes minimum at a pressure of 17 ± 1 pounds per square inch in a steam autoclave or sterilizer. The specimen shall be placed on its crown in the autoclave or sterilizer. At the end of the exposure period the following shall be accomplished:

a. The test specimen shall be immediately examined for evidence of blisters or delaminations, surface irregularities such as flat areas, bumps, or depressions and scalloped or rippled edges along the periphery of the liner. Evidence of any of the above conditions shall be considered as a test failure.

* b. The test specimen shall be conditioned at room temperature (70° to 80°F) for four hours minimum. After the conditioning period, the test specimen shall be inserted into an M-1 helmet (see MIL-H-1988) to determine conformance to the fit requirement specified in 3.6.3. Failure of the test specimen to pass the fit requirement shall be considered a test failure.

* 4.5.3 Ballistic evaluation. The ballistic evaluation shall be conducted in accordance with MIL-STD-662 except as indicated in 4.5.3.1. The fragment simulating projectile shall be the caliber 22, type 2, conforming to MIL-P-46593.

4.5.3.1 Ballistic test method. The liner shall be stored in the ballistic test chamber for not less than 24 hours prior to testing. It shall be tested in the region above the plane of the suspension rivet holes. The liner shall be rigidly mounted with the area to be subjected to impact normal to the line of fire. The witness plate shall be rigidly mounted inside the liner 2 inches behind the area of impact. The shots shall be a minimum distance of 1 1/2 inches from each other and shall be so dispersed as to be representative of the entire area above this plane. A shot shall be invalid if the shot causes delamination which extends into the delaminated area of the previous shot.

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4.5.3.2 Calculations. The V_{50} limit for the lot shall be reported as the average of all liners tested from the lot.

4.5.4 Water immersion test. The liners shall be immersed in tap water at 60° to 80°F for 16 hours minimum. At the expiration of this time, the liners shall be removed and allowed to air dry for 12 hours minimum or force air dry for 6 hours minimum under the same temperature conditions. The exterior coating shall be examined for evidence of softening, blistering or peeling.

4.5.5 Clip to stud attachment, type I liner. The suspension assembly shall be removed from the type I helmet liner. The suspension assembly shall be reinstalled as shown on Drawing 8-2-62 by hand only and without the use of tools or fixtures to hold the helmet liner in position. Failure to install any clip by hand shall constitute failure of the test.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Liners. Ten liners with suspension assembly in place, of one type only, shall be nested in a compact stack with a separator between each liner. The separator shall be made of embossed or indented paperboard having a thickness of not less than 0.016 inch prior to embossing or indentation or the equivalent. The separator may also be made of cushioning material conforming to PPP-C-1797. The separator shall be a disc having a diameter of approximately 17 inches and shall be slotted at 90° intervals around the circumference so that the disc will fold to the shape of the helmet liner.

5.1.1.2 Chin straps (for type II liners). Ten chin straps shall be compactly and neatly stacked to form a bundle. The bundle shall be securely tied with cotton tape or twine.

5.1.1.3 Headbands. When specified (see 6.2), 10 headbands, alternated end for end, shall be neatly stacked to form a bundle and securely tied at each end with cotton tape or twine. Prior to bundling, each assembled headband shall be adjusted to the smallest size and have all clips closed.

5.1.1.4 Neckbands. When specified (see 6.2), 10 neckbands shall be laid flat and evenly stacked to form a bundle. The extended adjusting tapes shall then be folded back over the tape webbing piece and the resultant bundle securely tied at each end with cotton tape or twine.

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* 5.1.2 Commercial. Liners and chin straps (when applicable), and headbands and neckbands when specified (see 6.2) shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Liners. Forty liners of one type only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-1, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed upright, two in length and two in width within a shipping container. The four stacks of liners shall be separated by the insertion of two "L" folded fiberboard sheets made from the same fiberboard material as the box liner. The prefolded fiberboard sheets shall measure 20 inches in length and 22 3/4 inches in width, with flutes and score running the width dimension. Inside dimensions of each shipping container shall approximate 22 1/2 inches in length, 19 inches in width, and 22 3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of the container specification.

* 5.2.1.1.1 Chin strap (for type II liners). Forty chin straps, preserved as specified in 5.1.1.2, shall be packed in an unfilled portion of each shipping container specified in 5.2.1.1 containing type II liners.

* 5.2.1.2 Headbands and neckbands. When specified (see 6.2), 40 headbands, preserved as specified in 5.1.1.3, and 40 neckbands, preserved as specified in 5.1.1.4, shall be packed together with the liners specified in 5.2.1.1. The items shall be placed in an unfilled portion of the shipping container.

5.2.2 Level B packing.

5.2.2.1 Liners. Forty liners of one type only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF, variety SW or SF, class domestic, grade 275 or PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed upright, two in length and two in width within a shipping

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container. The four stacks of liners shall be separated by the insertion of two "L" folded fiberboard sheets made from the same fiberboard material as the box liner. The prefolded fiberboard sheets shall measure 20 inches in length and 22 3/4 inches in width and flutes and score running the width dimension. Inside dimensions of each shipping container shall approximate 22 1/2 inches in length, 19 inches in width, and 22 3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents covered on the top and bottom with a sheet of 30-pound basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container document.

* 5.2.2.1.1 Chin straps (for type II liners). Forty chin straps, preserved as specified in 5.1.1.2, shall be packed in an unfilled portion of each shipping container specified in 5.2.2.1 containing type II liners.

5.2.2.1.2 Weather-resistant container. When specified (see 6.2), the fiberboard shipping container shall be grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the container document.

* 5.2.2.2 Headbands and neckbands. When specified (see 6.2), 40 headbands, preserved as specified in 5.1.1.3, and 40 neckbands, preserved as specified in 5.1.1.4, shall be packed together with the liners specified in 5.2.2.1. The items shall be placed in an unfilled portion of the shipping container.

* 5.2.3 Commercial packing. Liners and chin straps (when applicable), and when specified (see 6.2), headbands and neckbands, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), liners of one type only, chin straps (when applicable), headbands and neckbands (when specified), packed as specified in 5.2, shall be palletized in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The helmet liner (type I or type II) with the liner suspension assembly, headband and neckband attached is used with or without the Helmet, Ground Troop's - Parachutist's M-1, for ballistic protection and as Military head covering. The type II helmet liner with nonremovable suspension assembly is intended for use by parachutists because the chin strap and snap fasteners prevent loss of the helmet and helmet liner during descent.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type required (see 1.2).
c. When a first article is required (see 3.1.2, 4.3, and 6.3).
d. When production sample is not required (see 3.1.3).
e. When headbands and neckbands are required and whether they are to be furnished by the contractor or as Government Furnished Property (GFP) (see 3.2.3).
f. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
g. When headbands and neckbands are required (see 5.1.1.3, 5.1.1.4, 5.1.2, 5.2.1.2, 5.2.2.2 and 5.2.3).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1.2).
i. When palletization is required (see 5.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for inspection and approval of the first article.

* 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

6.5 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

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6.6 Head form. Information regarding standard AML wooden head form, size large, may be obtained from the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101. Head form may be obtained from M.A. Cumings Inc., 49 Bleecker St., New York, NY 10012.

6.7 Shell. Unless otherwise specified the term "shell" refers to the molded liner, trimmed, burnished, prepared for painting and drilled, without hardware and suspension assembly installed.

6.8 Preforms. The preform illustrated in Drawing 8-2-64 is offered for informational purposes only. This preform design has made acceptable liners. Any contractor electing to use them in part or in total does so with the knowledge that the Government is not liable for any expense or adverse results arising from them.

* 6.9 Laminating resin system formulation. The following resin system formulation has been used to make acceptable liners. Any contractor electing to use this formulation in part or in total does so with the knowledge that the Government is not liable for any expense or adverse results arising from its use.

Components	Parts by Weight	% of Total Solids
Polyvinyl Butyral (18% - 20% Hydroxyl)	868 (25% Solids in Ethanol)	47.2
Phenol Formaldehyde	100 (57% Solids in Ethanol)	12.4
Trimethylol Phenol	267 (60% Solids in Ethanol)	34.8
Phthalic Anhydride	25.6	5.6
Methanol	51.2	0
Totals	1311.8	100.0

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6.10 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, correction, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment

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Navy - AS
Air Force - 99

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-41800G	Liner, Ground Troop's - Parachutist's Helmet

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE SUBMISSION *(YYMMDD)*

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